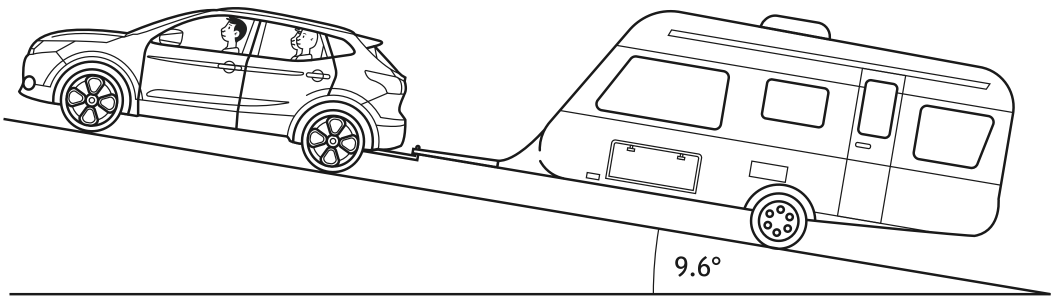
MARKS DO NOT WRITE IN THIS MARGIN

2. A car pulls a caravan up a slope at a constant speed of $4.0\ m\,s^{-1}$. The slope is at an angle of 9.6° to the horizontal.

The car and passengers have a total mass of 1650 kg.

The caravan has a mass of 1350 kg.



(a) (i) Determine the component of the total weight of the car, passengers, and caravan acting down the slope. **3**

Space for working and answer

(ii) The total frictional force acting on the car and caravan is 1800 N.

Determine the forward force produced by the car. **1**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

2. (continued)

(b) The car and caravan now accelerate uniformly up the slope for 250 s to a velocity of 9.5 m s^{-1}.

(i) Show that the acceleration of the car and caravan is 0.022 m s^{-2}. **2**

Space for working and answer

(ii) Determine the minimum forward force produced by the car while accelerating. **3**

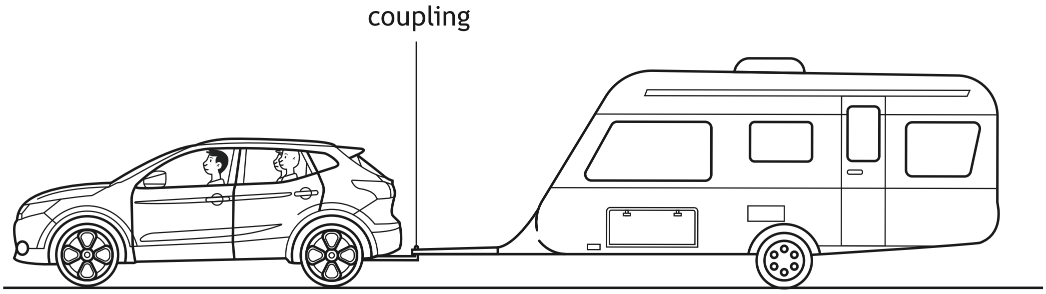
Space for working and answer

(iii) State one assumption you have made in your calculation for (b) (ii). **1**

[Turn over

2. (continued)

(c) Later in the journey the car and caravan are being driven along a straight, level road.



The car and caravan now accelerate at 0.16 m s^{-2}.

The frictional force acting on the car is 740 N.

The frictional force acting on the caravan is 1200 N.

Determine the tension in the coupling between the car and caravan. **3**

Space for working and answer